A Phenotype–Genotype Atlas for Common Recessive IRD Genes

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1. Macular involvement patterns — section 1

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 27804391, 36385862, 42209521.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 74182431, 85462980, 91862570.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 50904055, 70888425, 37631128.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 67089957, 78790752, 36385862.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 70888425, 23002814, 36385862.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 7531719, 54395632, 30838837.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 7531719, 67089957, 91862570.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 85462980, 50904055, 7531719.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 54395632, 5711300, 9978927.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 36385862, 11098043, 94334379.

2. ARL6 phenotype atlas — section 2

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 11098043, 91862570, 70888425.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 12799348, 75948090, 88330027.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Founder alleles complicate carrier-frequency interpretation in population isolates. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 30838837, 54395632, 6249541.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 91862570, 19353163, 94334379.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 67089957, 6249541, 28376016.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 70888425, 6249541, 63820668.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Founder alleles complicate carrier-frequency interpretation in population isolates. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 72884446, 86558084, 37631128.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 30838837, 75948090, 12799348.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 19353163, 75633113, 78790752.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 63820668, 42209521, 19353163.

3. Hearing involvement in USH2A — section 3

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 95658604, 50622942, 27804391.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 85462980, 12799348, 44462489.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 63820668, 6249541, 72884446.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 75633113, 85462980, 5711300.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 30838837, 92917383, 72884446.

Founder alleles complicate carrier-frequency interpretation in population isolates. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 85462980, 37631128, 37631128.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 50904055, 92917383, 85462980.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 92917383, 2101693, 91862570.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 91862570, 6249541, 50904055.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Founder alleles complicate carrier-frequency interpretation in population isolates. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 54395632, 86558084, 86558084.

Table D3. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	37458810	1059645	0.5772
2	72160023	3146665	0.3661
3	26550861	7686422	0.1147
4	90120021	8222178	0.5277
5	85879702	5268048	0.5968
6	82148427	6595795	0.5868
7	46797654	5915327	0.8448
8	82961016	2904531	0.8131

4. Polydactyly in ARL6/BBS3 — section 4

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 1268480, 75948090, 30838837.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 9978927, 19353163, 94334379.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 42209521, 67089957, 2101693.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 28376016, 37631128, 95658604.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 58238467, 36385862, 2101693.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 37631128, 88330027, 11098043.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 50622942, 70888425, 44462489.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 2101693, 4416102, 1268480.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 37631128, 42209521, 86558084.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 37631128, 12799348, 76072580.

5. Macular involvement patterns — section 5

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 78790752, 76072580, 95658604.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Founder alleles complicate carrier-frequency interpretation in population isolates. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 74182431, 27804391, 76072580.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 5711300, 5711300, 27804391.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 74182431, 70888425, 94334379.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 6249541, 88330027, 23002814.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 63888423, 7531719, 6249541.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 28376016, 78790752, 54395632.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 86558084, 91862570, 86558084.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 2101693, 27804391, 94334379.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 85462980, 72884446, 50904055.

6. RPE65 phenotype atlas — section 6

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 58238467, 94334379, 30838837.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 7531719, 63820668, 11098043.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 7531719, 85462980, 54395632.

Founder alleles complicate carrier-frequency interpretation in population isolates. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 24072653, 37631128, 54395632.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 75633113, 63888423, 44462489.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 63820668, 30838837, 94334379.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 95658604, 11098043, 37631128.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 86558084, 7531719, 85462980.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 28376016, 94334379, 75948090.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 30838837, 86558084, 24072653.

Table D6. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	81560345	8558370	0.1893
2	52679931	4968372	0.6459
3	32223815	8312838	0.3861
4	32457445	8391604	0.6378
5	9631069	8873176	0.8879
6	1719606	6068222	0.9331
7	17814442	6500068	0.8601
8	76386724	7070918	0.3657

7. Hearing involvement in USH2A — section 7

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 54395632, 12799348, 86558084.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 4416102, 9978927, 91862570.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 7531719, 30838837, 7531719.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Founder alleles complicate carrier-frequency interpretation in population isolates. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 78790752, 63888423, 30838837.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 91862570, 1268480, 72884446.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 54395632, 94334379, 54395632.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 11098043, 63820668, 94334379.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 70888425, 24072653, 63888423.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 4416102, 63820668, 74182431.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 12799348, 85462980, 27804391.

8. Hearing involvement in USH2A — section 8

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 54395632, 75948090, 50622942.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 50622942, 2101693, 9978927.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 70888425, 63888423, 27804391.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 44462489, 24072653, 50622942.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 28376016, 7531719, 2101693.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 27804391, 30838837, 95658604.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 42209521, 75948090, 23002814.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 54395632, 92917383, 54395632.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Founder alleles complicate carrier-frequency interpretation in population isolates. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 9978927, 88330027, 74182431.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 24072653, 9978927, 63820668.

9. Longitudinal ERG findings — section 9

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 6249541, 63888423, 78790752.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 5711300, 12799348, 28376016.

PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 44462489, 63888423, 58238467.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 11098043, 37631128, 72884446.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 37631128, 30838837, 72884446.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 19353163, 50622942, 86558084.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 6249541, 4416102, 74182431.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 37631128, 2101693, 2101693.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 1268480, 54395632, 74182431.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Founder alleles complicate carrier-frequency interpretation in population isolates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 7531719, 78790752, 78790752.

Table D9. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	34838319	1268076	0.0228
2	8753290	3327700	0.9017
3	59616344	2170763	0.0928
4	44599873	2065673	0.0436
5	61973691	859559	0.1687
6	77125433	7277158	0.8210
7	53115826	8333619	0.0304
8	91799285	7168234	0.1735

10. CERKL phenotype atlas — section 10

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 50622942, 95658604, 95658604.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 2101693, 75633113, 74182431.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 58238467, 24072653, 75948090.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 42209521, 88330027, 76072580.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Reference numeric values for this section: 67089957, 50904055, 28376016.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 75633113, 24072653, 76072580.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Reference numeric values for this section: 75948090, 70888425, 42209521.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 67089957, 75948090, 78790752.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 27804391, 58238467, 88330027.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 7531719, 74182431, 44462489.

11. Fundus autofluorescence patterns — section 11

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Reference numeric values for this section: 92917383, 95658604, 58238467.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 50904055, 23002814, 78790752.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 63820668, 6249541, 78790752.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Reference numeric values for this section: 11098043, 91862570, 88330027.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 78790752, 67089957, 36385862.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 27804391, 6249541, 27804391.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 88330027, 23002814, 92917383.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 7531719, 92917383, 37631128.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 76072580, 12799348, 85462980.

Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Reference numeric values for this section: 11098043, 9978927, 88330027.

12. DHDDS phenotype atlas — section 12

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Reference numeric values for this section: 27804391, 44462489, 11098043.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 28376016, 12799348, 85462980.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 63888423, 2101693, 7531719.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 19353163, 91862570, 11098043.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 9978927, 2101693, 42209521.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Reference numeric values for this section: 74182431, 6249541, 27804391.

Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 54395632, 91862570, 7531719.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Founder alleles complicate carrier-frequency interpretation in population isolates. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 86558084, 11098043, 5711300.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 75633113, 88330027, 12799348.

Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 36385862, 63820668, 67089957.

Table D12. Generic reference values (not relevant to any task).

Index	Metric A	Metric B	Metric C
1	59930944	758547	0.2638
2	62519022	4429132	0.2268
3	36145780	9476486	0.0625
4	93578748	1632400	0.0865
5	46242898	9094006	0.4142
6	79251743	3827030	0.5535
7	52056500	8777475	0.4188

Index	Metric A	Metric B	Metric C
8	71975527	7976231	0.5726

13. CERKL phenotype atlas — section 13

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Reference numeric values for this section: 4416102, 27804391, 24072653.

Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 11098043, 86558084, 58238467.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 58238467, 23002814, 19353163.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Reference numeric values for this section: 2101693, 6249541, 2101693.

CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 95658604, 78790752, 24072653.

DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 27804391, 78790752, 78790752.

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Reference numeric values for this section: 9978927, 12799348, 36385862.

Full-field electroretinography under ISCEV standards informs rod vs cone involvement. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Reference numeric values for this section: 70888425, 6249541, 23002814.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Ethnicity-matched control screening helps assess the rarity of novel variants in the relevant background. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. Reference numeric values for this section: 75948090, 74182431, 75633113.

RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. CERKL is associated predominantly with autosomal-recessive cone–rod dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 30838837, 70888425, 88330027.

14. Macular involvement patterns — section 14

Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Reference numeric values for this section: 28376016, 70888425, 9978927.

Functional validation in cell or animal models is the next step beyond statistical genetic evidence. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Founder alleles complicate carrier-frequency interpretation in population isolates. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Causal attribution from genetics alone is sometimes ambiguous when filtering returns multiple candidates. Reference numeric values for this section: 9978927, 88330027, 24072653.

ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Phenotype–genotype correlation is a recurring theme in clinical-genetics literature. Reference numeric values for this section: 30838837, 4416102, 6249541.

Segregation analysis with bidirectional Sanger sequencing supports candidate variant calls. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 58238467, 74182431, 75948090.

Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Functional validation in cell or animal models is the next step beyond statistical genetic evidence. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Founder alleles complicate carrier-frequency interpretation in population isolates. Reference numeric values for this section: 70888425, 63820668, 72884446.

Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Variant filtering pipelines typically apply minor-allele-frequency, impact, and zygosity criteria. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Reference numeric values for this section: 63820668, 88330027, 19353163.

Spectral-domain optical coherence tomography is widely used for retinal-layer phenotyping. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Full-field electroretinography under ISCEV standards informs rod vs cone involvement. ARL6/BBS3 alleles can cause syndromic Bardet–Biedl syndrome or non-syndromic recessive RP. Multi-generational pedigree sampling strengthens segregation evidence in recessive disorders. Reference numeric values for this section: 11098043, 12799348, 7531719.

USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. PDE6B loss-of-function alleles produce a rod-dominant photoreceptor death phenotype. Whole-exome sequencing has accelerated molecular diagnosis across recessive ophthalmic conditions. Population-frequency databases include 1000 Genomes, ExAC, and gnomAD.

Reference numeric values for this section: 27804391, 86558084, 44462489.

ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. Founder alleles complicate carrier-frequency interpretation in population isolates. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. DHDDS encodes dehydrodolichyl diphosphate synthase, with an Ashkenazi Jewish founder allele. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. Reference numeric values for this section: 54395632, 75633113, 94334379.

Population-frequency databases include 1000 Genomes, ExAC, and gnomAD. Inherited retinal degenerations (IRDs) are a heterogeneous group of monogenic disorders. ABCA4 alleles contribute to a wide spectrum from Stargardt disease to cone–rod dystrophy. RPE65 loss-of-function alleles cause Leber congenital amaurosis and early-onset severe retinal dystrophy. USH2A variants are associated with Usher syndrome type II and with non-syndromic recessive RP. Reference numeric values for this section: 70888425, 91862570, 36385862.

End of distractor document. No content above answers any specific pedigree-level question; this document is included to test whether downstream agents correctly select the relevant source paper.